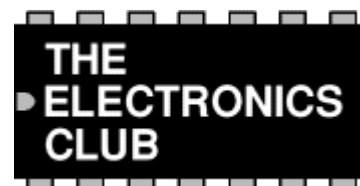


[Home](#) | [Map](#) | [Projects](#) | [Construction](#) | [Soldering](#) | [Study](#) | [Components](#) | [555](#) | [Symbols](#) | [FAQ](#) | [Links](#)

Ohm's Law



Next Page: [Power and Energy](#)

Also See: [Voltage and Current](#) | [Resistance](#) | [Resistors](#)

To make a current flow through a resistance there must be a voltage across that resistance. Ohm's Law shows the relationship between the voltage (V), current (I) and resistance (R). It can be written in three ways:

$$V = I \times R \quad \text{or} \quad I = \frac{V}{R} \quad \text{or} \quad R = \frac{V}{I}$$

where: V = voltage in volts (V) or: V = voltage in volts (V)
 I = current in amps (A) I = current in milliamps (mA)
 R = resistance in ohms (Ω) R = resistance in kilohms ($k\Omega$)

For most electronic circuits the amp is too large and the ohm is too small, so we often measure current in milliamps (mA) and resistance in kilohms ($k\Omega$). 1 mA = 0.001 A and 1 $k\Omega$ = 1000 Ω .

The Ohm's Law equations work if you use V, A and Ω , or if you use V, mA and $k\Omega$. You must not mix these sets of units in the equations so you may need to convert between mA and A or $k\Omega$ and Ω .

The VIR triangle

You can use the VIR triangle to help you remember the three versions of Ohm's Law.

Write down V, I and R in a triangle like the one in the yellow box on the right.

- To calculate **voltage, V**: put your finger over V, this leaves you with I R, so the equation is $V = I \times R$
- To calculate **current, I**: put your finger over I, this leaves you with V over R, so the equation is $I = \frac{V}{R}$
- To calculate **resistance, R**: put your finger over R, this leaves you with V over I, so the equation is $R = \frac{V}{I}$

V
I R

Ohm's Law
triangle

Ohm's Law Calculations

Use this method to guide you through calculations:

1. Write down the **Values**, converting units if necessary.
2. Select the **Equation** you need (use the VIR triangle).
3. Put the **Numbers** into the equation and calculate the answer.

V

It should be **Very Easy Now!**

I R

- 3 V is applied across a 6 Ω resistor, what is the current?
 - **Values:** $V = 3 \text{ V}$, $I = ?$, $R = 6 \Omega$
 - **Equation:** $I = \frac{V}{R}$
 - **Numbers:** Current, $I = \frac{3}{6} = 0.5 \text{ A}$

- A lamp connected to a 6 V battery passes a current of 60 mA, what is the lamp's resistance?
 - **Values:** $V = 6 \text{ V}$, $I = 60 \text{ mA}$, $R = ?$
 - **Equation:** $R = \frac{V}{I}$
 - **Numbers:** Resistance, $R = \frac{6}{60} = 0.1 \text{ k}\Omega = 100 \Omega$
(using mA for current means the calculation gives the resistance in k Ω)

- A 1.2 k Ω resistor passes a current of 0.2 A, what is the voltage across it?
 - **Values:** $V = ?$, $I = 0.2 \text{ A}$, $R = 1.2 \text{ k}\Omega = 1200 \Omega$
(1.2 k Ω is converted to 1200 Ω because A and k Ω must not be used together)
 - **Equation:** $V = I \times R$
 - **Numbers:** $V = 0.2 \times 1200 = 240 \text{ V}$

Next Page: [Power and Energy](#) | [Studying Electronics](#)

- [Electronics Club Home Page](#)
- [Site Map](#)
- [Example Projects](#)
- [Construction of Projects](#)
- [Soldering Guide](#)
- [Study Electronics](#)
- [Electronic Components](#)
- [555 Timer](#)
- [Circuit Symbols](#)
- [Frequently Asked Questions](#)
- [Links to other Electronics sites](#)



© John Hewes 2007, The Electronics Club, www.kpsec.freeuk.com